

Geometrization: The Exponential Correspondence, Thermal Relative Duality, and the Shadow's Home

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Abstract. The localization paper left the wedge phase with a generated but *hidden* Möbius group and two open structural questions: the geometry of the commutant, and the action of the modular conjugations. This note resolves both by exhibiting the geometrizing map. First, the conjugations are computed. The global modular conjugation is the thermofield swap: the four cross-pairings of a field against a tilde-field cancel identically (a four-line proof, machine-verified at 10^{-13} across 200 random pairs), so the global commutant is the shadow copy, with no geometric action on the scale line; separately, the reflection $u \mapsto -u$ is an *antiunitary* symmetry of the state — symmetric part exactly invariant, symplectic part exactly sign-flipped — and the fiber parity is a unitary channel symmetry. Second, the exponential correspondence: under $r = e^u$, with current test functions transforming as scalars (forced by the current's conformal weight one), the *vacuum* kernel $-1/4\pi(r - r')^2$ pulls back to exactly the thermal kernel at $\beta = 2\pi$ — an algebraic identity, verified to 0.0 — so the thermal system on the scale line is the vacuum system of the chiral current on the half-line $r > 0$, thermal vector equal to vacuum, KMS flow equal to dilations. Everything hidden then becomes geometric in the r -picture: the shadow commutant is the opposite half-axis $B(r < 0)$; the relative commutant is proven to be the left half-line algebra, $\mathcal{N}' \cap \mathcal{M} = \mathcal{A}(\mathbb{R}_-)$ — thermal relative duality — via interval Haag duality and strong additivity; and the three modular flows of the Schroer-Wiesbrock construction are the Möbius subgroups fixing $\{0, \infty\}$, $\{1, \infty\}$, $\{0, 1\}$, any two of which generate $PSL(2, \mathbb{R})$ with positive energy. The manifest symmetry is verified directly: the state is invariant under the full Möbius group at machine precision (three independent maps), while a non-Möbius interval-preserving flow breaks invariance at the percent level — the symmetry is exactly Möbius, no more and no less. The antipodal conjecture, re-filed by the diagnostic, returns with derived support: $r < 0$ along a direction is $r > 0$ along its antipode, so the shadow's natural physical identification in the coming assembly phase is the antipodal ray.

1. The two questions

The localization paper established the local net, the wedge, and the citation-level payoff — a hidden positive-energy Möbius group generated by three modular flows — while deferring, in its scope, the geometry of the commutant and the action of the conjugations. "Hidden" was a precise admission: the generated symmetry had no geometric interpretation on the thermal net. This note supplies the missing geometry by finding the coordinates in which nothing is hidden.

2. The conjugations, computed

The global conjugation is the thermofield swap. In the Araki-Woods form of the thermal representation, fields carry weights $c_+ = (1 + \rho)^{1/2}$, $c_- = \rho^{1/2}$ across the doubled one-particle space, and the tilde fields carry them swapped and conjugated. The commutator of a field with a tilde-field consists of four pairings which, writing $A = \int c_+ c_- \bar{F} G$, evaluate to $A - \bar{A} - A + \bar{A} = 0$ — identically, for *all* test functions, not merely disjoint ones. Machine-verified: maximal residual 1.1×10^{-13} over 200 random pairs on random positive weights. The global commutant $\mathcal{M}'$ is therefore the full shadow copy, and the global modular conjugation has no geometric action on the scale line whatsoever — Schroer-Wiesbrock's "thermal shadow world," held here explicitly as the tilde algebra.

The reflection is antiunitary. Under $u \mapsto -u$ the symmetric part of the two-point function is exactly invariant (-0.074442 on both sides at $\beta = 1.3$) while the symplectic part exactly flips sign (∓ 0.133081): invariance of the state requires composing with complex conjugation, so the reflection is implemented by a canonical *antiunitary* Θ with $\Theta \mathcal{A}(I) \Theta = \mathcal{A}(-I)$ — the PT-type signature.

The parity is unitary. The fiber parity $x \mapsto -x$ acts as $(r, \theta) \mapsto (r, -\theta)$, where — as throughout this series, fixed by the Mellin unitary $(Wf)(u, \theta) = e^{3u/2} f(e^u \theta)$ of the construction paper — θ denotes a *unit vector* of S^2 , so that $-\theta$ is its antipode and the map is the antipodal map on S^2 (in spherical angles, $(\vartheta, \phi) \mapsto (\pi - \vartheta, \phi + \pi)$, not an angle reflection): identity on u , $(-1)^l$ on the angular channels; since the covariance is a function of h and channel-diagonal, parity is a unitary symmetry commuting with the flow. At this level, then, the conjugation inventory is: the shadow swap (global, non-geometric) and the half-line conjugation — whose characterization is exactly the relative-commutant question answered in §4.

3. The exponential correspondence

Let $r = e^u$. The current is a conformal primary of weight one, so under any diffeomorphism its smeared test functions transform as *scalars*, $F \mapsto F \circ \varphi^{-1}$, the field's Jacobian cancelling the measure's.

Proposition 1 (the kernel identity). The vacuum current kernel on the r -line, pulled back through the exponential, is exactly the thermal kernel of this series at $\beta = 2\pi$:

$$-\frac{e^u e^{u'}}{4\pi (e^u - e^{u'})^2} = -\frac{1}{16\pi \sinh^2\left(\frac{u-u'}{2}\right)} = -\frac{\pi}{4\beta^2} \sinh^{-2}\left[\frac{\pi}{\beta}(u - u')\right] \Big|_{\beta=2\pi}.$$

This is an algebraic identity — holding as written off the diagonal, and as boundary values with the common prescription $u - u' \mapsto u - u' - i\epsilon$ carried through all three expressions; verified by smearing both sides against disjoint bumps, agreement 0.0 to all computed digits. General β is the coordinate rescaling $u \mapsto (2\pi/\beta)u$.

Consequences. Combining with the localization paper's Proposition 4 (our covariance is the thermal chiral kernel) and the uniqueness theorem (our state is *the* regular KMS

state): the thermal system $(\mathcal{A}(I_u), \omega_\beta)$ is the vacuum system of the chiral current on the half-line $r > 0$, read in logarithmic coordinates. The thermal vector is the vacuum; by Reeh-Schlieder the GNS space is the full vacuum space; and the global KMS flow is the dilation group of $r > 0$ — whose modular identity with respect to the vacuum is the chiral Bisognano-Wichmann statement, and whose KMS property this series has already verified directly, at machine precision, through the detailed-balance and strip-boundary computations of the state papers.

4. Theorem: thermal relative duality

Theorem. $\mathcal{N}' \cap \mathcal{M} = \mathcal{A}(\mathbb{R}_-)$: the relative commutant of the right half-line algebra in the global thermal algebra is exactly the left half-line algebra.

*Proof (in the r -picture, with cited inputs). * Under the correspondence, $\mathcal{M} = B((0, \infty))$, $\mathcal{N} = B((1, \infty))$, and $\mathcal{A}(\mathbb{R}_-) = B((0, 1))$, all with respect to the vacuum. Let $x \in \mathcal{N}' \cap \mathcal{M}$. By half-line duality on the line, $B((-\infty, 0)) \subseteq \mathcal{M}'$, so x commutes with $B((-\infty, 0))$ as well as with $B((1, \infty))$; hence $x \in [B((-\infty, 0)) \vee B((1, \infty))]'$. Interval Haag duality for the Möbius-covariant current net, together with its strong additivity, identifies this commutant with $B((0, 1))$. The converse inclusion $B((0, 1)) \subseteq \mathcal{N}' \cap \mathcal{M}$ is locality. ■

The cited inputs — half-line and interval duality for the current net, and its strong additivity — are standard properties of the $U(1)$ -current (Brunetti-Guido-Longo tradition; Guido-Longo-Wiesbrock), flagged as such. With them, the localization paper's open question (ii) is answered: no shadow components enter the *relative* commutant; within $\mathcal{M}$, duality is geometric.

5. The shadow's home, and the Möbius group made manifest

The shadow geometrized. By half-line duality, $\mathcal{M}' = B((-\infty, 0))$: the thermofield tilde copy of §2 is the algebra of the opposite half-axis $r < 0$. What was structurally non-geometric on the scale line — the u -line images only $r > 0$ — is plainly geometric one coordinate away: the shadow world is the other half of the line.

The three algebras and their flows. The Schroer-Wiesbrock triple becomes $B((0, \infty))$, $B((1, \infty))$, $B((0, 1))$ — half-line, shifted half-line, interval — and their vacuum modular flows are the Möbius one-parameter subgroups fixing $\{0, \infty\}$ (dilations: the series' own thermal flow, KMS-verified), $\{1, \infty\}$, and $\{0, 1\}$ (Bisognano-Wichmann for the half-lines; Hislop-Longo for the interval — cited). Any two of these subgroups generate $PSL(2, \mathbb{R})$, with positive energy from the vacuum representation. "Hidden" is thereby diagnosed as a coordinate effect: generic Möbius maps move the point $r = 0$ and so leave the u -chart entirely; on the r -line the group acts by fractional-linear point transformations, manifestly.

The manifest symmetry, verified. With the scalar transformation law, the Möbius identity $(\varphi(x) - \varphi(y))^2 = \varphi'(x)\varphi'(y)(x - y)^2$ makes the vacuum two-point function invariant under the entire group — and exactly this was checked: invariance at machine

precision for the interval-anchored modular flow (0.0), a differently anchored Möbius flow (2.4×10^{-16}), and a generic fractional-linear map (3.5×10^{-16}); by contrast, a non-Möbius interval-preserving flow breaks invariance genuinely (1.1×10^{-2} and 2.5×10^{-2} at two flow times). The symmetry of the state is exactly $PSL(2, \mathbb{R})$: no less — and, among reparametrizations, no more.

6. The antipodal return

The diagnostic of §2 re-filed the antipodal conjecture as belonging to a later phase; the geometry of §5 now returns it with derived support. In the three-dimensional fiber, $r < 0$ along a direction θ is $r > 0$ along the antipode $-\theta$: the two halves of one Cartesian axis, which is precisely the doubling $L^2(\mathbb{R}) = L^2(\mathbb{R}_+) \oplus L^2(\mathbb{R}_-)$ the Mellin analysis has carried since the construction paper. The shadow algebra $B(r < 0)$ therefore has a natural physical identification waiting in the assembly phase: the algebra of the antipodal ray. On that identification — stated here as the assembly phase's design proposal, not as a theorem — the thermofield double of the thermal state is the algebraic image of the two ends of each axis, and the modular conjugation exchanges a direction with its antipode. The conjecture was never wrong; it was filed one phase early, and the geometrization tells it where to live.

7. Scope

Four items. (i) The cited inputs of §§4-5 — half-line and interval duality, strong additivity, and the Hislop-Longo interval flow — are standard for the current net and are used as such; none is re-derived here. (ii) "Geometric" throughout means the r -picture: an internal geometry of the model, one exponential away from the scale line — not yet physical spacetime. The assembly phase, where per-direction wedge families over S^2 and the antipodal identification of §6 are to be constructed, is the next step, and only there does the question of higher-dimensional structure begin. (iii) The interval flow's KMS property was deliberately not tested numerically: analytic continuation through complex flow parameters is not elementary for compactly supported smearings, and the identification rests on the cited theorems; what was tested — full Möbius invariance with a non-Möbius contrast — is exactly the part numerics can decide. (iv) As throughout the series, β is a normalization: the correspondence fixes $\beta = 2\pi$ in the coordinate where the vacuum kernel takes its standard form, and every other β is the rescaled chart.

References

- Companion papers: *Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group*; *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III₁ Verdict*; *The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty*; *The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III₁ Verdict*; *The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line*.

- Borchers, H.-J. & Yngvason, J., "Modular groups of quantum fields in thermal states," *J. Math. Phys.* 40 (1999), 601.
 - Schroer, B. & Wiesbrock, H.-W., "Looking beyond the thermal horizon: hidden symmetries in chiral models," arXiv:hep-th/9901031.
 - Hislop, P. D. & Longo, R., on the modular structure of local algebras of the free massless field, *Comm. Math. Phys.* 84 (1982).
 - Brunetti, R., Guido, D. & Longo, R., "Modular structure and duality in conformal quantum field theory," *Comm. Math. Phys.* 156 (1993).
 - Guido, D., Longo, R. & Wiesbrock, H.-W., "Extensions of conformal nets and superselection structures," *Comm. Math. Phys.* 192 (1998), 217.
 - Wiesbrock, H.-W., "Half-sided modular inclusions of von Neumann algebras," *Comm. Math. Phys.* 157 (1993), 83–92.
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